Hugo Marquerie 11/11/2025

Desigualdad de Bessel

Teorema 1 (Desigualdad de Bessel). *Sea* $(H, \langle\cdot,\cdot\rangle)$ *un espacio prehilbert y* $\{u_n\}_{n\in\mathbb{N}}$ *un sistema ortonormal en* $H \implies \forall x \in H : \displaystyle\sum_{n=1}^{\infty} |\langle x, u_n\rangle|^2 \leqslant \|x\|^2$.

***Demostración*:** Calculamos la norma del vector $x - \sum_{n=1}^{N} \langle x, u_n\rangle u_n$:

$$
\begin{aligned}
0 &\leqslant \left\| x - \sum_{n=1}^{N} \langle x, u_n\rangle u_n \right\|^2 = \left\langle x - \sum_{n=1}^{N} \langle x, u_n\rangle u_n, x - \sum_{m=1}^{N} \langle x, u_m\rangle u_m \right\rangle \\
&= \|x\|^2 - \sum_{n=1}^{N} \overline{\langle x, u_n\rangle} \langle x, u_n\rangle - \sum_{m=1}^{N} \langle x, u_m\rangle \langle u_m, x\rangle + \sum_{n=1}^{N}\sum_{m=1}^{N} \overline{\langle x, u_n\rangle} \langle x, u_m\rangle \langle u_n, u_m\rangle \\
&= \|x\|^2 - 2\sum_{n=1}^{N} |\langle x, u_n\rangle|^2 + \sum_{n=1}^{N} |\langle x, u_n\rangle|^2 = \|x\|^2 - \sum_{n=1}^{N} |\langle x, u_n\rangle|^2 .
\end{aligned}
$$

Por tanto, $\forall N \in \mathbb{N} : \sum_{n=1}^{N} |\langle x, u_n\rangle|^2 \leqslant \|x\|^2$. Tomando límite cuando $N \to \infty$, se obtiene la desigualdad de Bessel. ■

Referenciado en

- `teo-identidad-plancherel`

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